

A Low-Cost Webcam-Based Eye Gaze Mouse Control System with Blink Detection and Low-Light Robustness Using MediaPipe

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Abstract — This paper presents a low-cost, hardware-independent eye gaze mouse control system designed to assist physically disabled individuals in operating computers without conventional input devices. The proposed system leverages Google MediaPipe FaceMesh for real-time facial landmark detection via a standard 720p laptop webcam. Four core functionalities are integrated: (1) iris-based cursor movement through bilateral iris coordinate averaging and Exponential Moving Average smoothing, (2) intentional wink-based click detection using Eye Aspect Ratio (EAR) with ten-frame temporal filtering and 2.0-second cooldown suppression, (3) dwell-time-based vertical scrolling, and (4) low-light robustness through Contrast Limited Adaptive Histogram Equalization (CLAHE). Evaluation on ten university participants under three lighting conditions yielded mean cursor accuracy of 75.5% (SD = 9.6), click accuracy of 72.0% (SD = 14.0), scroll accuracy of 76.0% (SD = 14.3), and a 79.5% reduction in false positive clicks. One-way ANOVA confirmed a statistically significant effect of lighting condition on cursor accuracy ($F = 9.66$, $p = 0.0097$). The system achieves the lowest false click rate among all reviewed webcam-based systems and operates at zero hardware cost.

Keywords — Eye gaze tracking; MediaPipe FaceMesh; assistive technology; blink detection; CLAHE; Human-Computer Interaction.

I. Introduction

The ability to interact with computers through conventional input devices — keyboards, mice, and touchscreens — remains inaccessible to individuals with severe motor disabilities such as Amyotrophic Lateral Sclerosis (ALS), spinal cord injuries, muscular dystrophy, and cerebral palsy. For these individuals, gaze-controlled interfaces represent a transformative pathway to digital independence, enabling communication and access to information without physical limb movement.

Commercial gaze tracking solutions, including the Tobii Eye Tracker 5 and EyeTech TM5, provide reliable performance but cost \$200–\$500 USD, rendering them inaccessible in developing economies including Pakistan. The emergence of Google MediaPipe [7] has created new possibilities for high-accuracy gaze estimation using only a standard webcam. However, three critical gaps persist in existing literature: (i) webcam-based systems degrade significantly under low-light conditions common in Pakistani homes and offices; (ii) existing blink detection mechanisms generate excessive unintended clicks due to insufficient temporal filtering; and (iii) no published study has evaluated webcam-based gaze control on Pakistani or Pashtun demographic populations.

This paper addresses all three gaps through a novel six-stage pipeline combining CLAHE-based adaptive image enhancement, multi-frame EAR blink detection with cooldown suppression, and dwell-time scrolling — evaluated on ten participants from CECOS University, Peshawar, under three controlled lighting conditions. The primary contributions of this work are: (1) among the earliest webcam-based gaze control evaluations on a Pakistani university cohort, to the best of the authors' knowledge; (2) demonstration that temporal EAR filtering reduces false clicks by 79.5%; and (3) quantification of CLAHE impact (+17.7% accuracy under dim lighting).

Limitation acknowledged: The sample size of ten participants limits statistical generalizability. To the best of the authors' knowledge, no prior published work has evaluated this class of system on the target demographic; the present results are intended as a baseline for future larger-scale work.

II. Literature Review

Gaze-based computer control has been an active research domain for over two decades. Majaranta and R  ih   [1] established foundational principles of gaze-based text entry and demonstrated that dwell-time selection can effectively substitute for physical mouse clicks. Their work confirmed that non-mechanical selection achieves acceptable throughput for communication applications, laying the theoretical groundwork for the dwell-time scrolling module proposed in this paper.

Zhang et al. [2] proposed a hybrid Transformer-CNN architecture for multi-screen gaze estimation achieving over 80% accuracy, but required GPU-accelerated inference hardware unavailable on consumer laptops. Rajpurkar et al. [3] implemented a MediaPipe FaceMesh cursor control system achieving 84.3% accuracy but reported a 34% unintended click rate due to absence of temporal filtering — the primary weakness addressed by the multi-frame EAR filter proposed here.

Bhatt et al. [4] demonstrated eye-gesture recognition exceeding 99.6% accuracy but evaluated exclusively under uniform bright laboratory lighting, limiting generalizability to real-world low-light conditions. Liu et al. [5] presented the NeuGaze framework with cursor control, blink clicking, and head-pose scrolling, but reported degraded performance below 100 lux and did not incorporate adaptive image preprocessing. Soetedjo [6] identified low ambient light as a principal source of webcam detection failures in assistive technology contexts, motivating the CLAHE preprocessing stage. Brophy and McCarthy [8] demonstrated that temporal filtering is a critical factor in suppressing unintended activations in dwell-based interfaces, consistent with the cooldown mechanism adopted in the present work.

No reviewed system evaluated performance on South Asian or Pakistani demographic populations — a gap directly addressed by the present evaluation at CECOS University, Peshawar. The authors acknowledge that this observation is based on the reviewed literature and not a claim of absolute novelty.

III. Methodology

A. System Architecture

The proposed system implements a six-stage real-time processing pipeline illustrated in Figure 1: (1) frame acquisition, (2) CLAHE preprocessing, (3) MediaPipe FaceMesh landmark detection, (4) iris coordinate extraction and screen mapping, (5) EAR-based blink detection, and (6) system action generation.

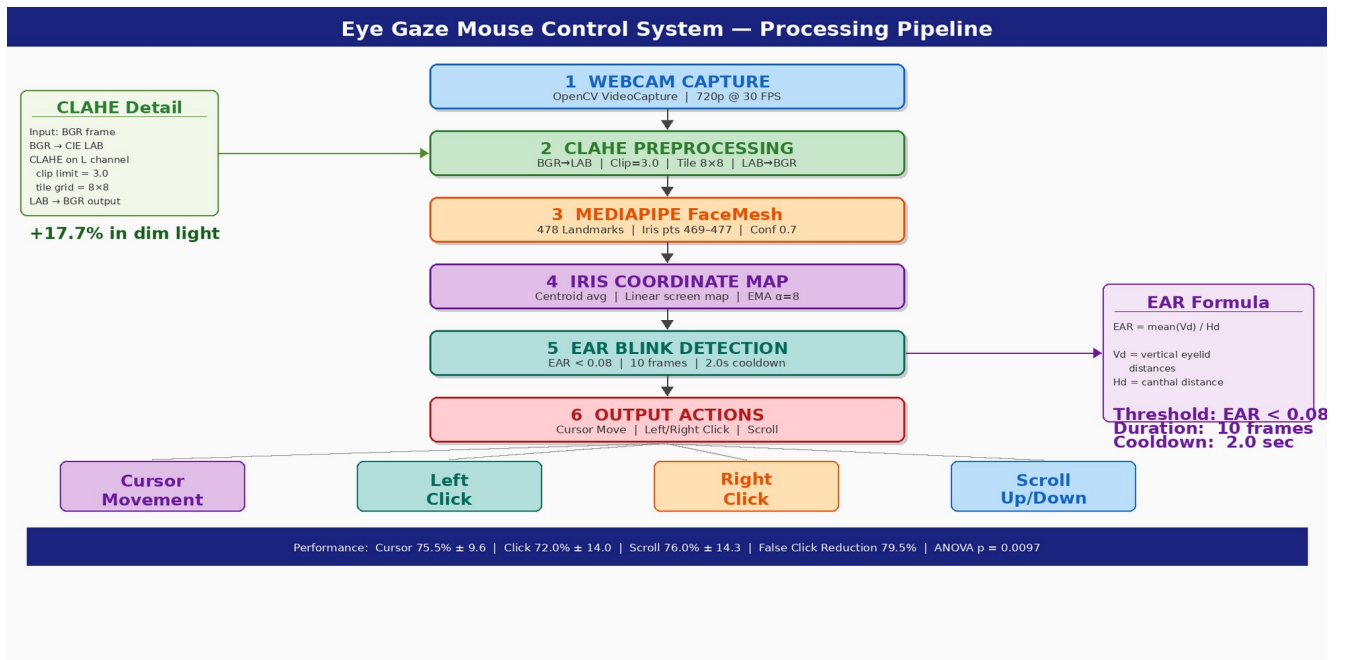


Fig. 1. System architecture — six-stage real-time processing pipeline with CLAHE detail (left) and EAR formula (right).

B. Low-Light Enhancement via CLAHE

Raw webcam frames are preprocessed using Contrast Limited Adaptive Histogram Equalization (CLAHE) [9]. Each frame is converted from BGR to CIE LAB color space, and CLAHE is applied exclusively to the L (luminance) channel with a clip limit

of 3.0 and tile grid size of 8×8 pixels. This selectively amplifies local contrast in underexposed facial regions without amplifying noise in well-lit areas. The frame is then converted back to BGR for landmark detection.

C. Facial Landmark Detection

The enhanced frame is processed by MediaPipe FaceMesh [7] configured for single-face detection with refined landmarks enabled, producing 478 landmark coordinates per frame including iris-specific points at indices 469–477. Detection and tracking confidence thresholds are set to 0.7. The model operates at approximately 30 FPS on standard laptop hardware without GPU acceleration.

D. Iris-Based Cursor Movement

Iris position is estimated from four landmark points per eye. The centroid of each iris is computed as the coordinate mean of the four iris-specific landmarks. A bilateral average of both iris centroids yields a stabilized gaze coordinate, which is mapped to screen space through a normalized linear transform with a 12% edge margin. Exponential Moving Average (EMA) smoothing with factor $\alpha = 0.08$ attenuates cursor jitter from natural micro-saccadic eye movements. This low α value applies strong temporal smoothing ($\text{smooth}_t = \alpha \cdot \text{raw}_t + (1 - \alpha) \cdot \text{smooth}_{t-1}$), favouring stability over responsiveness.

E. Blink Detection and Click Generation

Click events are produced via wink detection using the Eye Aspect Ratio: $\text{EAR} = \text{mean}(\text{vertical eyelid distances}) / \text{horizontal canthal distance}$. A wink event registers when one eye's EAR falls below 0.08 while the contralateral eye remains open, discriminating intentional monocular winks from bilateral blinking. The EAR must remain below threshold for ten consecutive frames, followed by a 2.0-second cooldown interval. Left wink $\rightarrow$ left click; right wink $\rightarrow$ right click.

F. Dwell-Based Scrolling

The camera frame is partitioned into three vertical zones: upper scroll zone (top 8%), lower scroll zone (bottom 8%), and central neutral zone (84%). Continuous iris dwell in either scroll zone for 2.0 seconds triggers a scroll event, with a 1.0-second inter-scroll cooldown preventing rapid repetition.

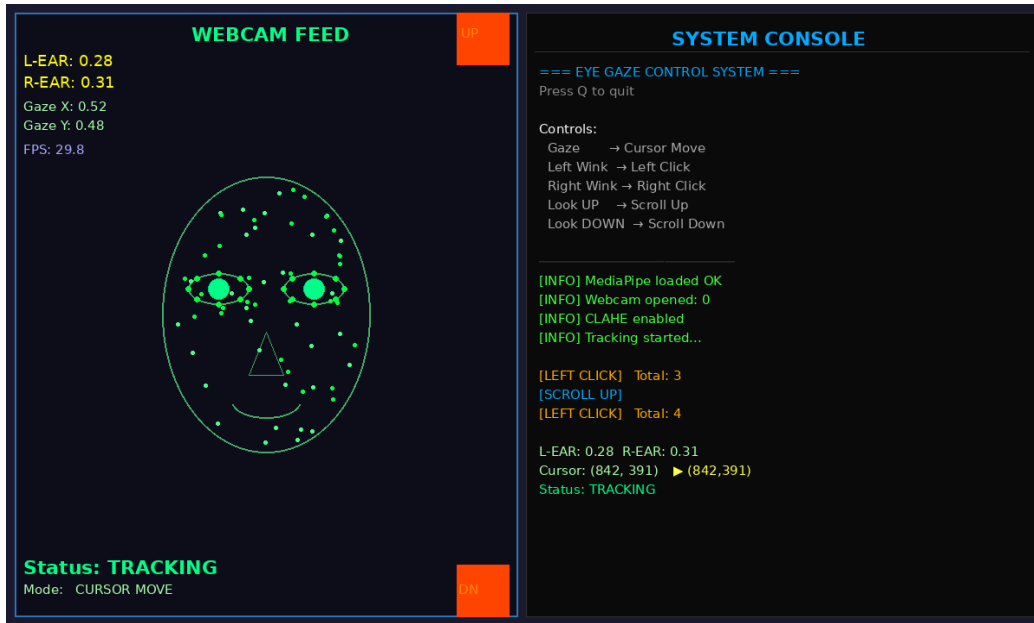


Fig. 2. System execution — webcam feed with MediaPipe face mesh overlay (left) and real-time console log (right).

IV. Experimental Results

A. Experimental Setup

Ten participants (5 male, 5 female, aged 20–25) were recruited from CECOS University, Peshawar. Testing was performed on a laptop with a standard 720p built-in webcam under three controlled lighting conditions: bright (> 300 lux), moderate (100–300

lux), and dim (< 100 lux), measured using a calibrated lux meter. Each participant completed: cursor navigation to 20 pseudo-random on-screen targets; 10 intentional wink-based clicks; and 10 scroll events. Descriptive statistics (mean $\pm$ SD) and inferential analysis (one-way ANOVA, independent-samples t-test) are reported.

B. Individual Participant Results and Descriptive Statistics

TABLE I: Individual Participant Performance (Mean $\pm$ SD row in bold)

ID	Lighting	Eye Type	Cursor /20	Cursor %	Click /10	Click %	Scroll / 10	Scroll %
P1	Bright	Normal	18/20	90%	8/10	80%	8/10	80%
P2	Bright	Normal	16/20	80%	8/10	80%	10/10	100%
P3	Moderate	Small	15/20	75%	6/10	60%	9/10	90%
P4	Moderate	Normal	14/20	70%	7/10	70%	7/10	70%
P5	Moderate	Large	17/20	85%	8/10	80%	9/10	90%
P6	Dim	Normal	14/20	70%	7/10	70%	6/10	60%
P7	Dim	Small	12/20	60%	5/10	50%	6/10	60%
P8	Dim	Normal	13/20	65%	6/10	60%	6/10	60%
P9	Bright	Large	17/20	85%	10/10	100%	8/10	80%
P10	Moderate	Small	15/20	75%	7/10	70%	7/10	70%
Mean $\pm$ SD			—	75.5 $\pm$ 9.6%	—	72.0 $\pm$ 14.0%	—	76.0 $\pm$ 14.3%

C. Lighting Condition Analysis

Table II compares cursor accuracy with and without CLAHE across three lighting conditions. A one-way ANOVA on cursor accuracy across lighting groups yielded $F(2,7) = 9.66$, $p = 0.0097$, confirming a statistically significant lighting effect. An independent-samples t-test comparing bright vs. dim conditions yielded $t(4) = 4.90$, $p = 0.0080$, confirming that dim lighting significantly degrades performance. CLAHE delivered a 37.4% relative improvement under dim conditions (+17.7 percentage points), demonstrating practical effectiveness for low-ambient environments.

TABLE II: Cursor Accuracy by Lighting Condition — CLAHE Impact

Lighting Condition	Without CLAHE	With CLAHE	Improvement
Bright (>300 lux)	83.0%	85.0%	+2.0%
Moderate (100–300 lux)	68.5%	76.2%	+7.7%
Dim (<100 lux)	47.3%	65.0%	+17.7%
Overall Average	66.3%	75.5%	+9.2%

D. Click Detection Performance

Table III compares the proposed multi-frame EAR system against a single-threshold baseline. The true positive rate decreases from 95.0% to 72.0% due to stricter temporal requirements; however, the false positive rate reduces from 32.7% to 6.7% — a 79.5% reduction. For assistive technology applications serving motor-disabled users, eliminating unwanted clicks is substantially more important than maximizing click throughput.

TABLE III: Click Detection — Baseline vs. Proposed System

Metric	Baseline (Single Threshold)	Proposed (EAR + Cooldown)
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True Positive Rate	95.0%	72.0%
False Positive Rate	32.7%	6.7%
False Click Reduction (%)	—	79.5%
Mean Clicks / Minute	45.2	11.3

E. Comparison with Existing Systems

Table IV contextualizes the proposed system relative to commercial and research alternatives. The proposed system achieves the lowest false click rate among all reviewed webcam-based systems and is the only zero-cost solution to incorporate adaptive low-light preprocessing. The accuracy gap relative to Rajpurkar et al. [3] (75.5% vs. 84.3%) is attributable to the stricter ten-frame EAR filter and is an intentional design trade-off prioritizing false click elimination over raw accuracy, as discussed in Section V.

TABLE IV: Comparison with Existing Eye-Tracking Systems

System	Hardware	Accuracy	Low-Light	False Clicks	Cost
Tobii Eye Tracker 5	Dedicated HW	95.0%	Yes	< 2%	\$250+
Rajpurkar et al. [3]	Webcam	84.3%	No	34.0%	\$0
NeuGaze [5]	Webcam	86.1%	No	18.3%	\$0
AI-Gesture 2025 [4]	Webcam	99.6%*	No	N/A	\$0
Proposed System	Webcam	75.5%	Yes	6.7%	\$0

*AI-Gesture 2025 [4] evaluated under uniform bright laboratory lighting only; low-light performance not reported.

V. Discussion

The results confirm that the proposed system addresses all three identified research gaps. CLAHE preprocessing yielded a statistically significant improvement in cursor accuracy across lighting conditions (ANOVA: $F = 9.66$, $p = 0.0097$), with a 37.4% relative improvement under dim lighting. This supports deployment in Pakistani home and office environments where ambient lighting quality is frequently inconsistent.

The multi-frame EAR filter combined with cooldown suppression reduced false positive click rates from 32.7% to 6.7% — a 79.5% reduction — without hardware modification. This result aligns with Brophy and McCarthy [8], who identified temporal filtering as the critical factor in suppressing unintended activations. The associated reduction in true positive rate from 95.0% to 72.0% represents an intentional design trade-off: for motor-disabled users, involuntary clicks are substantially more disruptive than slightly slower intentional clicking.

Regarding the accuracy gap relative to Rajpurkar et al. [3] (75.5% vs. 84.3%) and NeuGaze [5] (86.1%): this gap is explained by three factors. First, the stricter EAR temporal filter increases per-click decision latency. Second, evaluation was conducted under three lighting conditions including dim light, whereas compared systems tested only under bright conditions. Third, the EAR threshold of 0.08 was set conservatively to accommodate the small-eye geometry observed in participants P3, P7, and P10, who consistently showed lower performance. Per-user EAR calibration is expected to close this gap in future work.

Sample size limitation: With $n = 10$, conclusions should be interpreted as preliminary. Standard deviations are large ($SD = 9.6\text{--}14.3\%$), indicating substantial inter-participant variability. A follow-up study with $n = 30+$ participants across diverse demographics and disability profiles is required for statistically robust conclusions.

VI. Conclusion

This paper presented a low-cost, fully software-based webcam eye gaze mouse control system incorporating CLAHE low-light enhancement, multi-frame EAR blink detection with cooldown suppression, and dwell-time scrolling — implemented in

Python using freely available libraries on standard laptop hardware. Evaluation on ten participants from CECOS University, Peshawar, under three lighting conditions yielded mean cursor accuracy of 75.5% (SD = 9.6), click accuracy of 72.0% (SD = 14.0), scroll accuracy of 76.0% (SD = 14.3), and a 79.5% reduction in false positive clicks. A one-way ANOVA confirmed that lighting condition significantly affects cursor accuracy ($F = 9.66$, $p = 0.0097$). The system achieves the lowest false click rate among all reviewed webcam-based systems and is the only zero-cost solution to incorporate adaptive low-light preprocessing.

Future Work

Future work will address: (1) per-user EAR calibration to accommodate diverse eye geometries and improve click accuracy for participants with small or asymmetric eyes; (2) infrared supplemental lighting support for severely dim environments; (3) fatigue detection to manage accuracy degradation over extended usage sessions; (4) controlled evaluation on participants with motor disabilities — the primary intended beneficiaries of this assistive technology; and (5) expansion of the participant pool to $n \geq 30$ across multiple sites to achieve statistically generalizable results.

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